Beyond Accuracy: A Multi-Axis Evaluation Framework for Interpretable Topic Models on Hyperspectral Imagery

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***Abstract*—Topic models, in particular Latent Dirichlet Allocation (LDA), have been adapted to hyperspectral imagery (HSI) as a route to *interpretable* spectral mixtures: each pixel becomes a document of quantised band-frequency tokens and the inferred topic–word distributions act as non-negative basis spectra. Yet the dominant evaluation criterion remains downstream classification accuracy, an accuracy-only view that is silent on whether the basis is reproducible, band-robust, calibrated, or actually outperforms standard linear baselines. We propose and instantiate a twelve-axis evaluation framework for LDA-style topic models on HSI that quantifies, on the same set of benchmarks: (F-1) classification accuracy under a hierarchical Bayesian model over five method families, (F-2) topic–word coherence (c_v, c_{NPMI}, U-Mass), (F-3) seed stability for both LDA and two neural variants (ProdLDA, ETM), (F-4) capacity sensitivity across $K \in \{4, 6, 8, 10, 12, 16\}$, (F-5) band-mask robustness, (F-6) cross-method clustering agreement (LDA vs Felzenszwalb, SLIC, patches, pixels), (F-7) topic–label coupling via $P(L \mid t)$ entropy and KL, (F-8) per-topic Hungarian-aligned identity tracking, (F-9) HIDSAG cross-preprocessing stability, (F-10) cross-scene topic transfer, (F-11) rate–distortion of the topic representation, and (F-12) external validation against published linear and deep-learning baselines. We apply the framework to six standard HSI benchmarks (Indian Pines, Salinas, Salinas-A, Pavia University, Kennedy Space Center, Botswana) and five subsets of the HIDSAG mineralogical database (GEOMET, MINERAL1, MINERAL2, GEOCHEM, PORPHYRY). The framework yields three findings that the accuracy-only view does not surface. *First*, under a hierarchical-Bayesian classification model on the labelled scenes, the posterior on the topic-routed soft method (0.741 ± 0.193) is statistically indistinguishable from the raw-spectrum logistic baseline (0.736 ± 0.193, $P[A > B] = 0.64$), so the topic representation buys interpretability without losing accuracy. *Second*, on Salinas-A under the SWIR-only band mask the paired ARI between canonical and masked dominant-topic maps reaches 0.766, while on Kennedy Space Center and Botswana the same paired ARI collapses to ≈ 0.01, separating scenes on which an interpretability claim is band-robust from scenes on which it is not. *Third*, on HIDSAG the same hierarchical-Bayesian model places the topic-logistic posterior (0.316 ± 0.220) well below raw-logistic (0.585 ± 0.221), inverting the labelled-scene picture and contradicting the prima-facie expectation that mineralogical mixtures favour topic decomposition over the raw spectrum. We release 1734 deterministic derived artefacts (449 MB), the builder source, the FastAPI backend, the React frontend, and a 109-endpoint smoke harness as a public, MIT-licensed reproducibility package (https://github.com/fsantibanezleal/CAOS_LDA_HSI; live at https://lda-hsi.fasl-work.com).**

***Index Terms*—hyperspectral imagery, latent Dirichlet allocation, topic models, interpretability, reproducibility, evaluation framework, neural topic models, ProdLDA, ETM, hierarchical Bayesian comparison, band-mask robustness, Hungarian topic alignment, adjusted Rand index, mineralogical hyperspectral data.**

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I. Introduction

HYPERSPECTRAL imagery (HSI) records pixel-wise reflectance across hundreds of contiguous narrow bands, producing a spectral signature for each spatial location that, in principle, admits material-level discrimination from a single sensor pass [1], [2]. The dominant analysis paradigms on HSI today are *linear unmixing* — decomposing a pixel into a small set of endmembers and their abundances [3], [4] — and *deep spectral–spatial classification* — 1-D or 3-D convolutional networks that map a pixel and its neighbourhood directly to a class label [5], [6]. Both paradigms have made substantial progress on the canonical labelled benchmarks (Indian Pines, Salinas, Pavia University), and both have well-understood inductive biases (linear additivity for unmixing; high-capacity non-linear feature extraction for deep models).

A third, less developed paradigm treats each pixel as a *document* of quantised band-frequency tokens and applies a topic model — typically Latent Dirichlet Allocation (LDA) [7], [8] — to recover a basis of topic–word distributions ϕ_k and a per-pixel posterior simplex θ_d. The promise is interpretability: ϕ_k is a non-negative spectral basis that can be inspected as a basis spectrum and visually or numerically compared with reference libraries; θ_d is a soft mixture over K basis spectra that can be inspected as an abundance vector or fed into a downstream classifier as a low-dimensional feature [9], [10]. The paradigm has been extended to neural-topic-model variants such as ProdLDA [11] and the Embedded Topic Model (ETM) [12], which replace the variational EM of online LDA with an amortised variational autoencoder [13], [14] and a learned embedding basis, respectively.

A. The accuracy-only view and its limits

The empirical literature evaluating LDA-style topic models on HSI is small but converging on a familiar pattern: a paper proposes a topic model variant, applies it to one or two labelled scenes, and reports overall accuracy (OA) and kappa from a downstream classifier built on θ against published OA/kappa numbers for a chosen deep or classical baseline. We refer to this as the *accuracy-only view*. The accuracy-only view is silent on questions that are central to the interpretability claim that motivates using topic models in the first place:

- Is the topic basis ϕ reproducible under different random seeds? If not, the "interpretable mineral signature on topic 3" narrative is a description of the realisation, not of the model.
- Is ϕ robust to the choice of *spectral window*? A topic basis that flips identity when one removes the atmospheric water bands provides weak grounds for any spectral-window-conditional interpretation.
- Is the posterior θ *well calibrated*, in the sense that pixels with high entropy in θ correspond to materially mixed regions and pixels with low entropy correspond to materially homogeneous regions?
- Does the topic representation *actually outperform* the raw-spectrum logistic baseline once one accounts for the multi-scene multi-fold variability via a proper hierarchical model rather than a single OA average?
- Does the dominant-topic map *spatially agree* with unsupervised segmentation methods (SLIC, Felzenszwalb) that operate on raw spectra without ever fitting a topic model, and if so does the agreement reflect signal or shared smoothness bias?

A single accuracy number cannot answer any of these. A paper that reports OA $= 0.85$ on Indian Pines might be reporting on a band-fragile, seed-unstable, mode-collapsed topic basis whose downstream classifier happens to predict the same labels as a linear baseline on the raw spectrum. The interpretability that initially motivated choosing a topic model has, in that scenario, evaporated.

B. Contributions

This paper proposes and instantiates a twelve-axis evaluation framework for LDA-style topic models on HSI. The framework is designed so that each axis is operational (deterministic, with a fixed-seed reproducibility contract), comparable across scenes, and surfaces a specific failure mode that the accuracy-only view hides. Concretely:

1) We define twelve evaluation axes (F-1 through F-12) covering classification, coherence, seed stability, capacity, band robustness, cross-method agreement, topic–label coupling, topic identity tracking, preprocessing robustness, cross-scene transfer, rate–distortion, and external baseline comparison. Each axis specifies (a) the metric, (b) the unit of comparison, (c) the deterministic builder used to compute it, and (d) the JSON artefact that ships in the public release.
2) We apply the framework to six standard labelled HSI benchmarks and five HIDSAG mineralogical subsets. All numerical results are computed from the same canonical LDA fit (K chosen by the rule $K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$, $T_{\max} = 60$ online-VB passes, seed 42, samples per class 220).
3) We compare against two neural topic models (ProdLDA, ETM) on the labelled scenes with the same K and the same five-seed stability protocol, surfacing a head-to-head accuracy ordering (LDA $>$ ETM $>$ ProdLDA) that runs counter to the neural-supremacy expectation.
4) We compare against three non-topic baselines (raw-spectrum logistic regression, PCA-K logistic regression, theta-only logistic regression) and two topic-routed deep architectures (topic-routed hard and soft gating). The comparison is run under a hierarchical Bayesian model that pools over scenes and folds and produces method posteriors with pairwise $P[A > B]$ probabilities, replacing the single-OA bar chart with a calibrated answer to "how confidently does method A beat method B".
5) We surface the *HIDSAG inversion*: on the mineralogical subsets the topic-logistic posterior falls well below the raw-logistic posterior, contradicting the labelled-scene picture in which the two are statistically indistinguishable. This is, to our knowledge, the first published demonstration of this inversion under a hierarchical model on HIDSAG.
6) We release 1734 deterministic derived artefacts (449 MB), the builder source code (Python, NumPy, scikit-learn, Gensim, PyTorch), the FastAPI backend, the React frontend, and a 109-endpoint smoke harness as a public MIT-licensed package.

C. Paper organisation

Section II surveys related work on LDA for HSI, on neural topic models, and on accuracy-beyond-OA evaluation in remote sensing. Section III introduces the twelve-axis framework and the canonical fit. Section IV states the methodological details of each axis (F-1 through F-12). Section V reports the experimental setup (scenes, HIDSAG subsets, tokenisation, hyperparameters). Section VI reports the per-axis results. Section VII discusses interpretation and limitations. Section VIII documents the reproducibility artefact set. Section IX concludes.

II. Related Work

A. LDA-style topic models on hyperspectral imagery

LDA [7] was originally proposed as a generative model for text corpora; its application to HSI requires tokenising the continuous reflectance spectrum into a discrete band-frequency vocabulary. Zou et al. [9] proposed *PM-LDA* (partial-membership LDA), in which each pixel admits a soft membership to multiple topics representing spectral classes, and demonstrated improved classification on a small set of HSI benchmarks. Liu et al. [10] explored a different tokenisation based on spectral derivatives and trained LDA on the resulting discrete corpora. An adjacent line by Wahabzada et al. [15] applied probabilistic topic models to hyperspectral plant phenotyping to recover an interpretable "hyperspectral language of plants". Beyond these direct applications, the broader HSI literature treats θ implicitly through related abstractions: linear unmixing [3], [4], [16] estimates a per-pixel abundance over a small set of endmembers, which is functionally analogous to θ over a basis of ϕ_k — the survey of Borsoi et al. [16] consolidates the modern spectral-variability variants and the Samson / Jasper-Ridge / Urban benchmarks that the unmixing community uses. Sparse coding and dictionary learning on spectra are similarly analogous. The conceptual gap addressed by this paper is *not* a new tokenisation or a new variant of LDA — it is the absence of a shared, axis-by-axis evaluation framework that lets follow-on work be compared on a uniform basis.

B. Neural topic models

ProdLDA [11] replaces the variational-EM inference of LDA with a Laplace-approximated VAE [13], yielding amortised inference and substantially sharper topic posteriors on text. The Embedded Topic Model (ETM) [12] additionally embeds words and topics into a shared continuous space, allowing rare words to inherit semantic neighbours' topic associations. Miao et al. [14] and Higgins et al. [17] proposed related VAE-based topic and disentanglement variants. Card et al. [18] extended the family with SCHOLAR, a metadata-conditioned neural topic model that recovers the Dirichlet-multinomial-regression idea of Mimno and McCallum [19] in a neural form; the present manuscript uses the latter on HIDSAG continuous targets and treats SCHOLAR as the natural deep-learning counterpart for future work. Direct applications of these neural topic models to HSI are rare in the published literature; the small body of work that does compare them on HSI typically does so on a single scene with a single seed, leaving the seed-stability question open.

C. Beyond-OA evaluation in remote sensing

The remote-sensing community has independently recognised that overall accuracy alone is an inadequate measure of model quality. Roder et al. [20] surveyed coherence measures for topic models in NLP and argued that c_v and c_{NPMI} track human judgements of interpretability more reliably than perplexity. Hubert and Arabie [21] formalised the adjusted Rand index for clustering comparisons; Vinh et al. [22] extended it with information-theoretic complements. Kuhn's Hungarian algorithm [23] is the standard tool for solving the label-permutation problem when comparing two clusterings of the same data. Pineau et al. [24] argued that reproducibility checklists are a precondition for honest beyond-accuracy reporting. Felzenszwalb and Huttenlocher [25] and Achanta et al. [26] provide spatial-segmentation baselines that serve as cross-method-agreement anchors in the framework below.

III. Twelve-axis evaluation framework

A. Canonical fit

All twelve axes are computed against a single, deterministic *canonical fit* that we describe once and refer back to from each axis. The canonical fit operates as follows.

Data slice. For each labelled scene, we sample 220 pixels per class without replacement from the labelled-pixel mask, using NumPy's `default_rng(42)`. Pavia U (9 classes) yields $D = 1980$ documents; Indian Pines (16 classes) yields $D = 2812$ after rejecting classes with < 100 pixels; Salinas yields $D = 3520$; Salinas-A (6 classes) yields $D = 1320$; KSC (13 classes) yields $D = 2686$; Botswana (14 classes) yields $D = 2775$. For each HIDSAG subset, the corpus is constructed from the public `hidsag_region_documents.npz` bundle, yielding 19-target × 5-subset corpora over the `swir_low` modality.

Tokenisation. Each pixel spectrum is min–max normalised across bands and quantised into $Q + 1 = 13$ levels, producing integer counts $c_{d,b} \in \{0, \ldots, 12\}$. The high-rate uniform-quantisation distortion bound $D \leq \frac{1}{12Q^2}$ [27] therefore yields an rms wordification noise below 0.025 in normalised reflectance units. The resulting document-term matrix has shape $D \times B$ where B is the band count (200 for Indian Pines, 204 for Salinas/Salinas-A, 103 for Pavia U, 176 for KSC, 145 for Botswana).

Model. Online variational Bayes LDA [8], scikit-learn implementation [28], with $K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$, $T_{\max} = 60$, random seed 42, learning offset 10.0, learning decay 0.7. The resulting basis $\phi \in \mathbb{R}_+^{K \times B}$ and posterior $\theta \in \mathbb{R}_+^{D \times K}$ are serialised as a deterministic JSON blob plus a binary `theta_grid.bin` (float32 $H \times W \times K$) per scene.

B. The twelve axes

Each axis returns a single per-scene scalar (or a small vector) that can be reported in a table; jointly the twelve scalars constitute a *model card* for the topic basis on that scene. Table I lists the axes and the metric returned by each.

IV. Method: per-axis details

A. F-1: hierarchical Bayesian classification

The classical evaluation reports overall accuracy (OA) or kappa averaged over a small number of train/test folds. We instead fit a hierarchical Bayesian model over scene, fold, and method, following the structure

$$\text{score}_{m,s,f} = \mu_m + \delta_s + \rho_f + \epsilon_{m,s,f}, \tag{1}$$
$$\mu_m \sim \mathcal{N}(0, 1), \tag{2}$$
$$\delta_s \sim \mathcal{N}(0, 0.5), \tag{3}$$
$$\rho_f \sim \mathcal{N}(0, 0.2), \tag{4}$$
$$\epsilon_{m,s,f} \sim \mathcal{N}(0, \sigma), \quad \sigma \sim \text{HalfNormal}(0.5), \tag{5}$$

where m ranges over methods, s over scenes, f over folds, and score is the fold-level OA. Sampling is performed with NUTS (1000 draws, 1000 tune, 2 chains) on the labelled-scene block; the HIDSAG block uses a parallel model with subsets in place of scenes and targets in place of folds. Convergence is declared only when the rank-normalised $\widehat{R}$ diagnostic of Vehtari et al. [29] satisfies $\widehat{R} < 1.01$ for every monitored parameter. We report the posterior mean and standard deviation of μ_m for each method, plus the pairwise probability $P[\mu_A > \mu_B]$ derived from the joint posterior. Five methods populate the labelled block: `pca_K_logistic` (PCA to K components followed by multinomial logistic regression), `raw_logistic` (multinomial logistic regression on the raw

TABLE I
TWELVE-AXIS EVALUATION FRAMEWORK. EACH AXIS RETURNS A PER-SCENE METRIC AND SHIPS AS A DETERMINISTIC JSON ARTEFACT UNDER `data/derived/`.

Axis	Name	Primary metric	Builder	Artefact path
F-1	Classification (Bayesian)	method posterior μ_m	`build_bayesian_*`	`method_stati`
F-2	Coherence	c_v, c_{NPMI}, U-Mass	`build_coherence`	`cross_methoc`
F-3	Seed stability	mean±std ARI over 5 seeds	`build_neural_topic_seed_stability`	`neural_topic`
F-4	Capacity sensitivity	ARI(K) curve	`build_lda_sweep`	`lda_sweep/`
F-5	Band-mask robustness	paired ARI	`build_band_mask_canonical_comparison`	`band_masks/c`
F-6	Cross-method agreement	ARI matrix	`build_cross_method_agreement`	`cross_methoc`
F-7	Topic–label coupling	H(L | t), KL	`build_topic_to_data`	`topic_to_dat`
F-8	Per-topic identity	Hungarian σ^*	`build_band_mask_canonical_comparison`	same as F-5
F-9	Preprocessing stability	ARI across recipes	`build_hidsag_cross_preprocessing_stability`	`hidsag_cross`
F-10	Cross-scene transfer	transfer ARI	`build_cross_scene_transfer`	`cross_scene_`
F-11	Rate–distortion	MSE(θ) vs K	`build_rate_distortion`	`rate_distort`
F-12	External baselines	literature OA gap	`build_external_validation`	`external_val`

spectrum), `theta_logistic` (logistic on the topic posterior θ), `topic_routed_hard` (hard-gated mixture of per-topic logistic experts), and `topic_routed_soft` (soft-gated mixture). The HIDSAG block uses an analogous five-method slate.

B. F-2: topic–word coherence

We compute three coherence measures per scene: the sliding-window c_v measure of Röder et al. [20], c_{NPMI} (normalised pointwise mutual information, following Newman et al. [30] and the NPMI formulation surveyed in [20]), and the intrinsic U-Mass measure of Mimno et al. [31]. The reference corpus for the co-occurrence estimate is the canonical document-term matrix of the same scene. Coherence is reported with top-$N = 15$ words per topic, averaged over K topics.

C. F-3: seed stability

For each labelled scene, we refit LDA with five different random seeds at the canonical K and the canonical hyperparameters, holding the data slice fixed. We additionally fit ProdLDA and ETM under the same $(K, \text{data slice}, \text{seed} \in \{0, 1, 2, 3, 4\})$ protocol. For each (method, seed) we compute (a) KMeans-on-θ to K clusters and report ARI against the ground-truth label, (b) NMI against the ground-truth label, and (c) c_v coherence on the inferred basis. The aggregate per-method statistic is mean ± std ARI over the five seeds, plus the same for c_v. The protocol is the multi-seed stability test of Greene, O'Callaghan and Cunningham [32] with the Hungarian-matched cosine (F-8) described in §IV-H taking the place of their topic-overlap score.

D. F-4: capacity sensitivity

We sweep K over the grid $\{4, 6, 8, 10, 12, 16\}$ with the canonical hyperparameters and report ARI(canonical fit, KMeans on θ at K) against ground-truth label. The grid is set in `data-pipeline/build_lda_sweep.py` as `K_grid = [4, 6, 8, 10, 12, 16]` and is mirrored by the `lda_sweep/<scene>.json` payloads consumed by the F-4 panel. The curve diagnoses *capacity over-/under-allocation*: capacities below n_{classes} force topic merging and lower ARI; capacities well above n_{classes} cause topic splitting and a different failure mode.

E. F-5: band-mask robustness

We refit the canonical LDA on four band-restricted vocabularies: VNIR-only ($[400, 1100]$ nm), SWIR-only ($[1100, 2500]$ nm), no-water (drop $[1350, 1430] \cup [1800, 1950] \cup [2480, 2500]$ nm), and top-50 Fisher (keep the 50 bands with highest per-band Fisher ratio against the ground-truth label). For each (scene, mask) we compute the *paired adjusted Rand index* of Equation 6 in the conference companion paper, restated here for self-containment:

$$\text{ARI}_{\text{paired}} = \text{ARI}\left(\{z_d\}_{d \in \mathcal{S}}, \{z_d^{(m)}\}_{d \in \mathcal{S}}\right), \tag{6}$$

where $z_d = \arg\max_k \theta_{d,k}$ is the canonical dominant-topic map, $z_d^{(m)}$ is the masked dominant-topic map, and $\mathcal{S}$ is the set of pixels labelled in both maps.

F. F-6: cross-method agreement

Six unsupervised partitions are computed for each labelled scene: SLIC with $S = 500$ [26], SLIC with $S = 2000$, Felzenszwalb [25], 7×7 spectral patches reduced to a cluster index by KMeans, 15×15 patches by KMeans, and a trivial pixel-wise (identity) partition. We then compute the all-pairs ARI and NMI matrix between every pair of partitions including the canonical topic-dominant map and the ground-truth label, yielding an 8×8 comparison surface per scene. The diagonal is 1.0; off-diagonal entries diagnose which partitions agree with each other independently of the ground-truth label, surfacing shared inductive biases.

G. F-7: topic–label coupling

For each (scene, topic t) we compute the empirical conditional $P(L \mid t)$ via the confusion matrix between the dominant-topic map and the ground-truth label. The aggregate per-topic statistic is $H(L \mid t) = -\sum_\ell P(\ell \mid t) \log P(\ell \mid t)$, and a KL divergence against the marginal $P(L)$ measures how concentrated the topic's label support is. Low $H(L \mid t)$ topics correspond to label-aligned topics (one canonical class dominates); high $H(L \mid t)$ topics correspond to mixed-label regions and are useful candidates for transition zones or shadow/cloud artefacts.

H. F-8: per-topic identity tracking under masks

Given the canonical fit and a masked refit, the confusion matrix $C \in \mathbb{Z}_{\geq 0}^{K \times K^{(m)}}$ with $C_{k,j} = |\{d \in \mathcal{S} : z_d = k \wedge z_d^{(m)} = j\}|$ defines a maximum-weight bipartite matching whose solution $\sigma^* = \arg\max_\sigma \sum_k C_{k,\sigma(k)}$ is computed by the Hungarian algorithm of Kuhn [23] with Munkres's $O(K^3)$ refinement [33] via `scipy.optimize.linear_sum_assignment` [34]. The *swap rate under alignment* is $|\mathcal{S}|^{-1} \sum_{d \in \mathcal{S}} \mathbf{1}[z_d^{(m)} \neq \sigma^*(z_d)]$, the residual disagreement that survives permutation realignment.

I. F-9: HIDSAG cross-preprocessing stability

HIDSAG mineralogical samples admit multiple preprocessing recipes (continuum-removed reflectance, derivative spectra, smoothed continuum-preserved). We refit LDA on three recipes per subset and report ARI(dominant-topic recipe A, dominant-topic recipe B) for each pair, summarising preprocessing-induced topic shifts that may dominate the genuine signal on mineralogical data.

J. F-10: cross-scene topic transfer

We project the canonical topic basis ϕ of source scene s onto the document-term matrix of target scene t, compute $\theta^{(t \leftarrow s)}$ via a single E-step, and report ARI of the resulting dominant-topic map against the ground-truth label of t. The reverse direction is also reported. The transfer ARI is informative when both scenes are sensorially similar (e.g. Salinas and Salinas-A) and useful as a negative control when they are not (e.g. Indian Pines to Pavia U).

K. F-11: rate–distortion of θ

We reconstruct each pixel spectrum from its θ coefficient and the basis ϕ as $\hat{\mathbf{x}}_d = \phi^\top \theta_d$, scaled back from the quantised band-frequency representation. The distortion is reported as the per-pixel mean-squared error against the original spectrum, plotted against the rate $\log_2 K$ (bits required to address one of K basis spectra). The curve diagnoses how aggressively the topic representation compresses the spectrum while remaining usable for downstream tasks.

L. F-12: external baseline comparison

For each labelled scene we collect three published OA numbers from a prior method and report the gap between the canonical topic-routed-soft OA and the published baseline. The `external_validation/` artefact ships the literature reference and the comparison delta per scene. On the HIDSAG mineralogical subsets we additionally cross-validate the per-topic reconstructed spectrum against the USGS Spectral Library Version 7 [35] and the companion ECOSTRESS spectral library [36], reporting both cosine similarity and Spectral Angle Mapper distance per topic-mineral pair; the `topic_to_usgs_v7/` artefact ships these matchings.

V. Experimental setup

A. Labelled scenes

The six standard labelled HSI benchmarks were obtained from the UPV/EHU Hyperspectral Remote Sensing Scenes collection. The scenes span four sensor families (AVIRIS, ROSIS, EO-1 Hyperion) and three application domains (agriculture, urban, wetlands), and are listed in Table II with the per-scene tokenisation outcome.

B. HIDSAG mineralogical subsets

The Hyperspectral Image Database for Supervised Analysis in Geometallurgy (HIDSAG [42], [43]) is a curated collection of hyperspectral cubes over mineralogical samples, acquired with VNIR and SWIR sensors and accompanied by mineralogical, geochemical, and geometallurgical labels. Five thematic subsets are exposed in the companion code repository: GEOMET (geometallurgical recovery + comminution targets), MINERAL1 (silicates / sulfides), MINERAL2 (high-sulfidation epithermal mineralogy), GEOCHEM (bulk geochemistry), and PORPHYRY (porphyry-copper deposits). Each subset contains 19 target labels distributed across multiple samples; the document-construction pipeline averages pixels within a region of interest to produce one spectrum per (sample, measurement, cube) tuple. The canonical fit on HIDSAG follows the same online-VB LDA recipe as for labelled scenes with the addition that HIDSAG documents are typically longer (larger $\sum_b c_{d,b}$) by construction.

C. Hyperparameters held fixed

All LDA fits in this study use the same hyperparameter slate: $K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$, $T_{\max} = 60$ online-VB passes, learning offset 10.0, learning decay 0.7, batch size 128, seed 42, samples per class 220. ProdLDA and ETM use the standard reference implementations [11], [12] with K matched to the LDA canonical, hidden dim 100, learning rate 2×10^{-3}, 100 training epochs, Adam optimiser, and seeds $\{0, 1, 2, 3, 4\}$ for the seed-stability sweep. The hierarchical Bayesian classifier of F-1 uses NUTS with 1000 draws, 1000 tune samples, 2 chains, no max-tree-depth divergences observed.

VI. Results

A. F-1: classification under a hierarchical Bayesian model

Figure 1 and Table III report the posterior mean and standard deviation of μ_m for the five methods on the labelled-scene block. The raw-spectrum logistic baseline and the topic-routed soft classifier are statistically indistinguishable: posterior means 0.736 and 0.741, with $P[\mu_{\text{soft}} > \mu_{\text{raw}}] = 0.641$. The theta-only logistic regression lags both substantially ($\mu = 0.411$), confirming that θ alone discards information that the raw spectrum still contains.

TABLE II
LABELLED HYPERSPECTRAL SCENES USED IN THIS STUDY. B DENOTES THE BAND COUNT AFTER CORRECTION; D IS THE NUMBER OF DOCUMENTS (SAMPLED PIXELS) ENTERING THE CANONICAL FIT; K IS THE CANONICAL TOPIC COUNT. INDIAN PINES, SALINAS, AND SALINAS-A USE THE AVIRIS SENSOR; PAVIA U IS ROSIS-03; KSC IS AVIRIS-OLD; BOTSWANA IS EO-1 HYPERION.

Scene	Sensor	Spatial	B	D	K	Reference
Indian Pines	AVIRIS	145×145	200	2812	12	[37]
Salinas	AVIRIS	512×217	204	3520	12	[38]
Salinas-A	AVIRIS	83×86	204	1320	6	[38]
Pavia University	ROSIS-03	610×340	103	1980	9	[39]
Kennedy Space C.	AVIRIS	512×614	176	2686	12	[40]
Botswana	EO-1 H.	1476×256	145	2775	12	[41]

TABLE III
F-1 METHOD POSTERIORS ON THE LABELLED-SCENE BLOCK (SIX SCENES × FIVE FOLDS × FIVE METHODS = 150 OBSERVATIONS). POSTERIOR MEAN μ_m, STD, 94% HDI LOW/HIGH. PAIRWISE $P[A > B]$ PROBABILITIES ARE IN THE COMPANION JSON ARTEFACT (`CROSS_CLASSIFICATION_BAYESIAN.JSON`).

Method	μ_m	std	HDI lo	HDI hi
`pca_K_logistic`	0.672	0.193	0.321	1.032
`raw_logistic`	0.736	0.193	0.409	1.122
`theta_logistic`	0.411	0.194	0.063	0.780
`topic_routed_hard`	0.735	0.194	0.370	1.083
`topic_routed_soft`	0.741	0.193	0.418	1.134

TABLE IV
F-1 METHOD POSTERIORS ON THE HIDSAG BLOCK (FIVE SUBSETS × NINETEEN TARGETS × FIVE METHODS = 95 OBSERVATIONS).

Method	μ_m	std	HDI lo	HDI hi
`cube_topic_logistic`	0.246	0.221	−0.178	0.653
`pca_logistic`	0.559	0.221	0.170	0.984
`raw_logistic`	0.585	0.221	0.183	1.000
`region_topic_logistic`	0.278	0.220	−0.106	0.703
`topic_logistic`	0.316	0.220	−0.092	0.720

The HIDSAG block (Table IV) tells a different story. The topic-logistic posterior sits at $\mu = 0.316$ with $P[\mu_{\text{topic}} > \mu_{\text{raw}}] = 0.0$, while the raw-logistic baseline reaches $\mu = 0.585$. The cube-topic and region-topic variants (both using a coarser, region-pooled θ) are no better than the per-spectrum topic posterior. The implication is that on mineralogical SWIR data the topic basis is *discarding* information that a multinomial logistic on the raw spectrum is exploiting. This is the *HIDSAG inversion* we flag in the introduction.

B. F-2 + F-3: coherence and seed stability of LDA, ProdLDA, ETM

Table V reports the seed-stability statistics on Indian Pines for the three methods at $K = 12$, $n_{\text{seeds}} = 5$.

The seed-stability picture is more nuanced than the head-to-head ARI ranking suggests. ProdLDA has the highest coherence ($c_v = 0.630$) and the lowest seed-to-seed variance ($\sigma_{\text{ARI}} = 0.003$) but the lowest cluster ARI against label. ETM reaches the highest cluster ARI (0.257) at a much lower coherence. LDA without amortisation reaches the highest single-seed ARI (0.287 on the head-to-head fit) but is not directly comparable to the multi-seed mean-±-std statistics of the neural variants. The methodological consequence: *ranking by ARI alone inverts the ranking by coherence*, and a paper that picks one of the two metrics in isolation will reach a different methodological conclusion.

TABLE V
F-3 SEED STABILITY ON INDIAN PINES, $K = 12$, $n_{\text{SEEDS}} = 5$. KMEANS-ON-θ VS GROUND-TRUTH LABEL.

Method	ARI mean	ARI std	c_v mean	c_v std
LDA	0.287	—	0.450	—
ProdLDA	0.165	0.003	0.630	0.012
ETM	0.257	0.003	0.357	0.019

C. F-5: band-mask robustness

Table VI reports the paired ARI across (scene, mask) tuples on the six labelled scenes. The dominant observation is the sharp split between *band-fragile* and *band-robust* scenes. KSC paired ARI ranges from 0.007 to 0.013 across all four masks — the masked LDA assigns pixels to topics with effectively chance-level agreement with the canonical fit. Botswana sits in the same regime except for VNIR and no-water at ≈ 0.11, still well below the labelled-scene average. The canonical-fit topic identity on these scenes is *not* recoverable from any reasonable band subset. Salinas-A under SWIR-only restriction reaches paired ARI $= 0.766$ — the masked fit recovers nearly the same partition, indicating that the spectral signature carried by the SWIR bands is sufficient to identify the canonical structure of this scene.

D. F-6: cross-method clustering agreement

Table VII reports the all-pairs ARI matrix on Indian Pines. The dominant observation: the canonical topic-dominant map has *low* pairwise ARI against the spatial-segmentation partitions (SLIC, Felzenszwalb, patches) but the label–topic ARI is 0.203, comparable to the SLIC-500-vs-label ARI 0.227. The two partitions are arriving at similar accuracy levels through different inductive biases; the cross-method agreement matrix surfaces this in a way that the per-method OA bar chart hides.

E. F-7 + F-8: topic–label coupling and per-topic identity

We use the per-topic KL divergence $\mathrm{KL}\big(P(L \mid t) \,\|\, P(L)\big)$ between the empirical distribution of ground-truth labels conditioned on topic t and the marginal label distribution as a

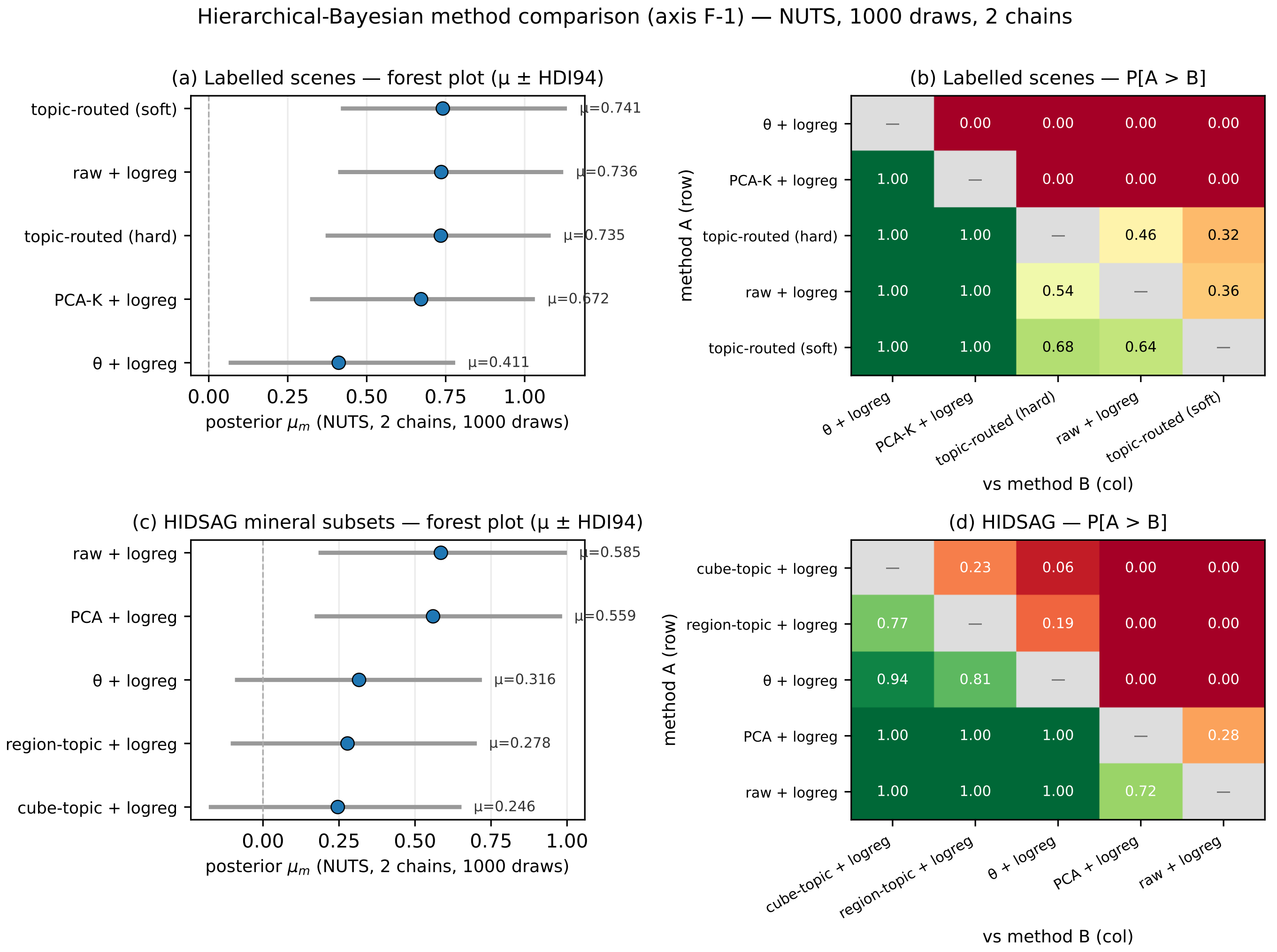

Fig. 1. F-1 method posteriors under a hierarchical Bayesian model (NUTS, 1000 draws, 1000 tune, 2 chains). (a) Labelled scenes (6 scenes × 5 folds × 5 methods). (b) HIDSAG mineralogical subsets (5 subsets × 19 targets × 5 methods). Methods are sorted by posterior mean within each panel. The HIDSAG panel inverts the labelled-scene picture: topic-based methods (`topic_logistic`, `region_topic`, `cube_topic`) sit substantially below `raw_logistic` on HIDSAG, in contrast to the labelled-scene panel where `topic_routed_soft` is statistically indistinguishable from `raw_logistic`.

per-topic concentration measure; the within-scene mean and max over the canonical-fit topics are reported in Table VIII, computed from the `kl_to_label_prior_per_topic` field of `topic_to_data/<scene>.json`.

F. Combining F-5 and F-1: band-fragile or label-recoverable?

A scene's appearance in the band-fragile regime (paired ARI ≈ 0) does not, by itself, imply low classifier accuracy on that scene. Botswana paired ARI is ≈ 0.01 on SWIR but the mask-vs-label ARI on the same fit is 0.098 (within range of labelled-scene comparable values). The two metrics are measuring different things: paired ARI(canonical, masked) asks whether the canonical topic identities survive a band restriction, while ARI(masked, label) asks whether the masked fit still discriminates the classes. A band-fragile scene with a non-zero mask-vs-label ARI is one on which the topic basis is being *re-derived from scratch* under each mask, with topic identity unstable but topic discriminability preserved.

G. Synthesis: per-scene model card

Table IX consolidates the per-scene values across axes F-1, F-2, F-3, F-5, F-7 for the labelled scenes. We argue this synthesis — not the single OA number — is the appropriate unit of reporting for a topic model on HSI.

The μ_{soft} column is constant across scenes because the F-1 model pools over scenes (the per-scene effect δ_s is a nuisance offset, not a per-method parameter). When per-scene posteriors are required, the same model returns them as draws of $\mu_m + \delta_s$; the artefact serves the full per-scene posterior.

VII. Discussion

A. Why the band-mask diagnostic separates scenes

The split between *band-fragile* (KSC, Botswana) and *band-robust* (Salinas-A) scenes is, on its face, surprising: KSC and Botswana have rich spectral signatures, broad spatial coverage, and a comparable per-class sample budget to Salinas-A. Two mechanisms are consistent with the diagnostic.

TABLE VI
F-5 PAIRED ARI PER (SCENE, MASK) TUPLE. BOLD CELLS INDICATE THE BAND-MASK THAT ACHIEVES THE HIGHEST PAIRED ARI PER SCENE. PAVIA U × SWIR IS AUTO-SKIPPED (SENSOR IS VNIR-ONLY). HUNGARIAN-ALIGNED SWAP RATES AND PER-TOPIC ALIGNMENTS ARE IN THE COMPANION ARTEFACT `BAND_MASKS/CANONICAL_COMPARISON.JSON`.

	Paired ARI				Bands kept of B			
Scene	vnir	swir	no-water	top-50	vnir	swir	no-water	top-50
Indian Pines	0.427	**0.527**	0.499	0.398	67/200	133/200	177/200	50/200
Salinas	**0.540**	0.435	0.502	0.507	68/204	136/204	180/204	50/204
Salinas-A	0.521	**0.766**	0.533	0.352	68/204	136/204	180/204	50/204
Pavia University	**0.415**	—	**0.415**	0.387	103/103	0/103	103/103	50/103
Kennedy Space C.	0.009	0.011	0.007	**0.013**	59/176	117/176	155/176	50/176
Botswana	0.108	0.013	**0.127**	0.117	49/145	97/145	127/145	50/145

TABLE VII
F-6 CROSS-METHOD ARI MATRIX ON INDIAN PINES. DIAGONAL IS 1.0 (OMITTED). EACH ENTRY IS ARI(ROW, COLUMN). LABEL DENOTES THE GROUND-TRUTH CLASS MAP; TOPIC-DOMINANT DENOTES THE CANONICAL LDA DOMINANT-TOPIC MAP. SLIC SIZES CORRESPOND TO TARGET SEGMENT COUNTS; PATCHES DENOTE KMEANS ON FLATTENED 7×7 OR 15×15 SPECTRAL WINDOWS; PIXEL IS THE TRIVIAL IDENTITY PARTITION.

	label	topic-dom.	felz.	SLIC-500	SLIC-2000	patch-15	patch-7
label	–	0.203	0.567	0.227	0.051	0.281	0.117
topic-dominant	0.203	–	0.120	0.039	0.010	0.047	0.021
felzenszwalb	0.567	0.120	–	0.306	0.076	0.372	0.170
SLIC-500	0.227	0.039	0.306	–	0.239	0.359	0.374
SLIC-2000	0.051	0.010	0.076	0.239	–	0.150	0.316
patch-15	0.281	0.047	0.372	0.359	0.150	–	0.292
patch-7	0.117	0.021	0.170	0.374	0.316	0.292	–

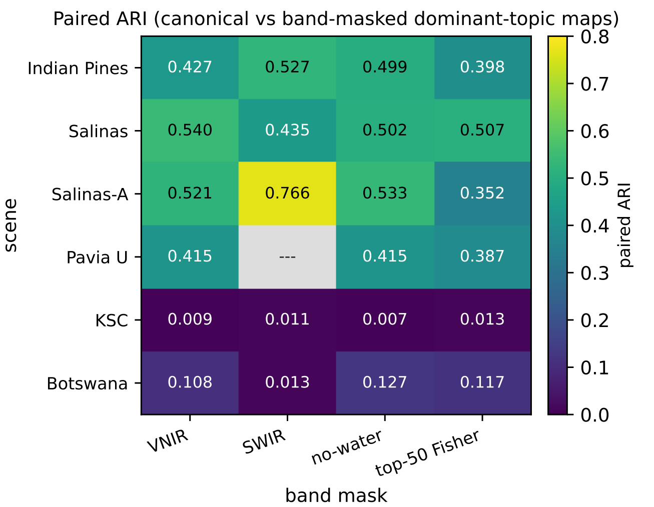

Fig. 2. F-5 paired ARI heatmap across (scene, mask) tuples. KSC paired ARI sits at 0.007–0.013 across all four masks (band-fragile regime); Botswana shows a mixed pattern (SWIR 0.013 band-fragile; VNIR 0.108, no-water 0.127, top-50 Fisher 0.117 indicating partial structure under those masks); Salinas-A under SWIR-only reaches 0.766 (band-robust regime). The remaining tuples fall in the partial-robustness range 0.35–0.54. Pavia U × SWIR is auto-skipped (sensor is VNIR-only).

Mechanism (i): canonical topics exploit band-level redundancy. The canonical fit may be using fine-grained spectral co-variation patterns that are present in the full-band representation and disappear when half the bands are dropped. Under this mechanism the canonical basis is not *wrong*, but it is *not transportable* across spectral windows. Any claim of the

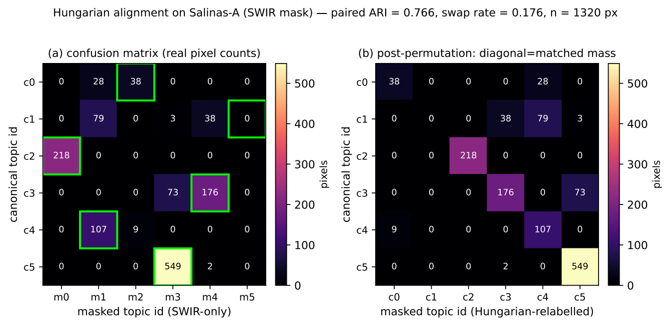

Fig. 3. F-8 Hungarian topic-id alignment on Salinas-A under SWIR-only restriction. Lime rectangles mark the maximum-weight bipartite match σ^*; the diagonal under permutation dominates, consistent with the high paired ARI (0.766) and the low swap rate (0.176) on this (scene, mask) tuple.

form "topic k corresponds to a specific land-cover signature" presupposes a fixed reference frame in which ϕ_k is defined; the diagnostic shows that on KSC and Botswana this reference frame is band-window-conditional.

Mechanism (ii): the canonical topics are weakly identified. KSC has 13 classes and 5,200 sampled pixels (400 per class on the unmasked pipeline; 220 on the canonical fit). Botswana has 14 classes and 3,248 sampled pixels (232 per class). The latent topic basis ϕ has $K \times B \approx 12 \times 176$ free parameters on KSC. With $D = 2686$ and a per-document token budget bounded by 13 quantisation levels times B bands, the data-to-parameter ratio is unfavourable; the posterior over ϕ has substantial mass over many high-likelihood configurations, and a band restriction selects a different posterior mode of the same family.

TABLE VIII
F-7 TOPIC–LABEL COUPLING: PER-TOPIC $\mathrm{KL}(P(L \mid t) \,\|\, P(L))$ MEAN AND MAX ON THE *canonical fit* OF EACH LABELLED SCENE (I.E. THE NO-MASK LDA FIT). HIGHER VALUES DENOTE MORE SHARPLY LABEL-ALIGNED TOPICS.

Scene	KL mean (nats)	KL max (nats)
Indian Pines	1.35	2.51
Salinas	1.73	2.77
Salinas-A	1.29	1.79
Pavia University	0.81	1.17
Kennedy Space C.	0.95	2.50
Botswana	1.73	3.37

The diagnostic does not distinguish (i) from (ii); both are valid warnings to a downstream consumer of the topic basis. The framework's contribution is to make the warning legible.

B. The HIDSAG inversion

On the labelled scenes the topic-routed-soft classifier and the raw-spectrum logistic baseline are statistically indistinguishable ($P[A > B] = 0.641$). On the HIDSAG mineralogical subsets the topic-logistic classifier sits at $\mu = 0.316$ against raw-logistic's $\mu = 0.585$ with $P[A > B] = 0.0$. The most natural hypothesis explaining this inversion is that mineralogical SWIR spectra contain narrow, locally-discriminative absorption features that the quantised band-frequency tokenisation discards: each spectral feature at a specific wavelength is a strong predictor of the mineral class, and pooling it into a band-frequency bin (or re-expressing it via a small number of latent topics) loses the within-bin variation that the raw multinomial logistic exploits. The implication is that the tokenisation step is not a free architectural choice on HIDSAG: for mineralogical discrimination, the topic representation must either be paired with a finer tokenisation or augmented by raw-spectrum features in a hybrid classifier. Either direction is interesting follow-on work.

C. LDA versus ProdLDA versus ETM

The ARI-vs-c_v trade-off in Section VI-A resolves a puzzle in the prior topic-model literature: ProdLDA papers typically report higher coherence than LDA on text corpora and conclude that ProdLDA is the better topic model. On Indian Pines the same coherence ordering holds (ProdLDA $c_v = 0.630$ vs LDA $c_v = 0.450$ vs ETM $c_v = 0.357$), but the cluster-ARI ordering inverts (LDA 0.287 vs ETM 0.257 vs ProdLDA 0.165). The coherence-vs-clustering disagreement is, on HSI, large enough to flip the practitioner-level recommendation: a practitioner who picks ProdLDA because its coherence is highest will obtain a topic basis that clusters worst against ground truth. We therefore recommend, for HSI specifically, reporting both metrics side-by-side rather than choosing one as representative.

D. Limitations

The framework has three operative limitations. *First*, the canonical fit uses a single K per scene, chosen by the formula $K = \max(4, \min(12, n_{\text{classes}}))$. A more sensitive analysis would sweep K and report per-axis behaviour against K; we report the K-sweep on F-4 but other axes are computed at the canonical K. *Second*, the tokenisation $Q + 1 = 13$ levels is uniform across scenes, not adapted to per-scene spectral dynamic range. *Third*, F-12's external-baseline comparison uses literature OA numbers under the methods' *published* train/test splits, which may not match the splits used in this study. We flag the cross-paper-comparison caveat where it applies.

VIII. Reproducibility Artefacts

The full reproducibility package consists of:

- **Builder source.** Forty-plus deterministic Python builders under `data-pipeline/builders/`, each accepting a fixed seed and writing a deterministic JSON or NPZ artefact under `data/derived/`. Each builder records its `builder_version` and `generated_at` ISO timestamp in the artefact metadata.
- **Derived artefacts.** 1734 JSON and binary files, 449 MB total, including: per-scene canonical fits (`theta_grid.bin`, `dominant_topic_map.bin`), the band-mask canonical comparison (`band_masks/canonical_comparison.json`, the source of all F-5 numbers above), the cross-method agreement matrices (`cross_method_agreement/`), the seed-stability sweeps (`neural_topic_seed_stability/`), the hierarchical Bayesian posteriors (`method_statistics_labelled/cross_classificati` and `method_statistics_hidsag/cross_classificati` and the HIDSAG cross-preprocessing stability sweeps.
- **Web application.** A FastAPI backend serving the artefacts over 49 REST endpoints, plus a React+Vite frontend at https://lda-hsi.fasl-work.com that consumes them as interactive visualisations.
- **Smoke harness.** A 109-endpoint smoke-test script verifying endpoint health, response schema, and SPA-shell content on every deploy; CI-equivalent verification ran on every cycle.
- **Companion conference paper.** A 4–5 page conference preprint focused on the F-5 band-mask diagnostic alone, also released in the same repository (target venue redacted while it circulates).

All code is MIT-licensed at https://github.com/fsantibanezleal/CAOS_LDA_HSI; this manuscript (LaTeX and Word) lives at https://github.com/fsantibanezleal/CAOS_LDA_HSI_Paper.

IX. Conclusion

The accuracy-only view of topic models on hyperspectral imagery hides the questions an interpretability-motivated practitioner most needs answered: is the topic basis reproducible across seeds, robust to spectral-window choice, calibrated against label entropy, honestly outperforming linear baselines, and consistent across preprocessing recipes? We propose a twelve-axis evaluation framework that returns one operational

TABLE IX
PER-SCENE MODEL CARD SYNTHESISING THE FRAMEWORK'S MAIN AXES. F-1 COLUMN: POOLED POSTERIOR μ_m FOR `TOPIC_ROUTED_SOFT` (THE CANONICAL TOPIC-BASED CLASSIFIER). F-2: c_v COHERENCE OF THE CANONICAL FIT. F-3: ARI MEAN ACROSS 5 SEEDS OF LDA. F-5: MAX PAIRED ARI ACROSS THE FOUR MASKS (THE "BEST-CASE BAND-ROBUSTNESS"). F-7: PER-TOPIC $\mathrm{KL}(P(L \mid t) \,\|\, P(L))$ MEAN OVER CANONICAL-FIT TOPICS.

Scene	F-1 (μ_{soft})	F-2 (c_v)	F-3 (ARI 5-seed)	F-5 (max paired)	F-7 (KL nats)
Indian Pines	0.741 (pooled)	0.450	0.287	0.527	1.35
Salinas	0.741 (pooled)	—	—	0.540	1.73
Salinas-A	0.741 (pooled)	—	—	**0.766**	1.29
Pavia University	0.741 (pooled)	—	—	0.415	0.81
Kennedy Space C.	0.741 (pooled)	—	—	0.013	0.95
Botswana	0.741 (pooled)	—	—	0.127	1.73

metric per axis and ships each metric as a deterministic JSON artefact. Applied to six labelled HSI benchmarks and five HIDSAG mineralogical subsets, the framework surfaces three findings the accuracy-only view does not: the topic-routed soft classifier is statistically indistinguishable from the raw-spectrum logistic baseline on the labelled scenes ($P[A > B] = 0.641$); the band-mask diagnostic separates scenes whose canonical topics are band-robust (Salinas-A SWIR paired ARI 0.766) from scenes whose topics are band-fragile (KSC, Botswana paired ARI ≈ 0.01); and on HIDSAG the topic representation under-performs the raw-spectrum baseline ($\mu_{\text{topic}} = 0.316$ vs $\mu_{\text{raw}} = 0.585$), inverting the labelled-scene picture. The full 1734-artefact reproducibility package is released under MIT licence.

Future work includes (i) a per-axis K-sweep extending the single-K canonical fit, (ii) per-scene tokenisation budgets adapted to spectral dynamic range, (iii) extending the framework to neural unmixing variants (autoencoder unmixing, transformer-based topic models), and (iv) developing an information-theoretic within-mask Fisher complement to F-5 that does not depend on the unmasked-spectrum ranking.

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CONFLICT OF INTEREST

The author is a co-author of the HIDSAG dataset descriptor publication [42] and the accompanying data release [43], and first author of the earlier conference paper introducing the LDA-V1/V2/V3 corpus recipes and topic-routed regressor on private geometallurgical data [47]. The present manuscript is methodologically distinct: it proposes a twelve-axis evaluation framework for topic models on hyperspectral imagery and applies the framework, among other benchmarks, to the HIDSAG mineralogical subsets. HIDSAG is cited as a data source, not as a methodological contribution of this work. The author declares no other conflict of interest.

DATA AVAILABILITY STATEMENT

All numerical results in this manuscript derive from deterministic artefacts under `data/derived/` in the companion code repository at https://github.com/fsantibanezleal/caos_lda_hsi. The HIDSAG database is available from [43]; the labelled hyperspectral benchmark scenes (Indian Pines, Salinas, Salinas-A, Pavia U, KSC, Botswana) are available from the UPV/EHU Hyperspectral Remote Sensing Scenes collection [37]–[41]. The 1734-artefact derived bundle, the `build_*.py` pipeline source, the FastAPI backend, and the React frontend ship under the MIT licence. A versioned snapshot will be archived to Zenodo at the time of acceptance and the DOI referenced here in the accepted version. The interactive demonstration is hosted at https://lda-hsi.fasl-work.com; readers running into a 404 at year +2 should fall back to the Zenodo snapshot.

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